1 Your documents — SciFact

5,183 DOCS · NO QUERIES · NO LABELS

- Effect of losartan on... 1
- Myelin oligodendrocyte...
- Alcohol Sales and Risk of... 2
- Experimental and clinical...
- Vitamin D insufficiency and...
- Chronic pancreatitis is... 3
- Gene set enrichment...

Task: Given a scientific claim, retrieve documents that support or refute the claim

2 Write queries — DeepSeek-V4-Flash

```
Task: Given a scientific claim, retrieve documents that
support or refute the claim
Documents:
  [1] Effect of losartan on ambulatory…
  [2] Alcohol Sales and Risk of Serious…
  [3] Chronic pancreatitis is essential…
Write ONE query the task describes, not a restatement
of any shown document.
```

What non-genetic factors can trigger cancer development in the presence of oncogenic mutations?

✓ 3–4,000 characters ✓ contains words × 100 queries

3 Retrieve — 28 models, run as in mteb

+23

Qwen3-Embedding-8B — top 10 · similarity

1	Cancer-related inflammation The...	0.592
2	Chronic pancreatitis is essential...	0.584
3	Evolution of metastasis revealed...	0.576
4	Hallmarks of Cancer: The Next...	0.575
5	Trans-ancestry mutational...	0.564

... ranks 6–10

The pool for this query: every document any model returned, **87** in all.

4 Judge — JEV-27B

For every document in the pool: a probability for each grade 0–5. The grade is their mean.

Qwen3-Embedding-8B, rank 2
Chronic pancreatitis is essential for...



grade **4.26**

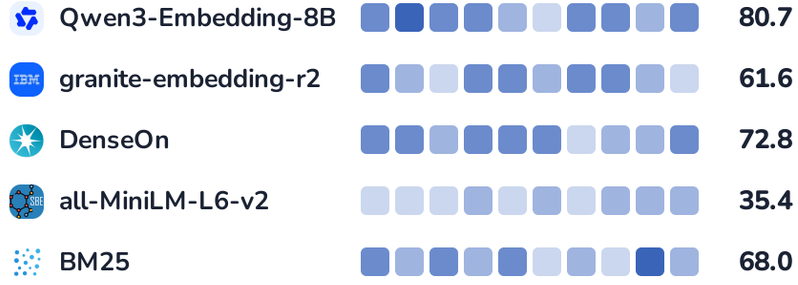
all-MiniLM-L6-v2, rank 1
Colorectal Cancers from Distinct...



grade **1.06**

5 Score — nDCG@10 with the judge's grades

each model's top 10, by grade 0 1 2 3 4 5 nDCG@10



nDCG@10 = DCG of the model's 10 grades ÷ DCG of the 10 best grades in the pool. A model's score is its mean over the **100 queries**.

6 Rank the models — SciFact

1	granite-embedding-r2	89.4
2	DenseOn	89.4
3	Qwen3-Embedding-8B	89.1
⋮		
25	all-MiniLM-L6-v2	80.2
26	BM25	79.4

ρ = 0.81 Spearman with MTEB's ranking from human relevance labels on SciFact